MIL-C-17/19A
9 October 1979
SUPERSEDING
MIL-C-17/19
7 September 1955

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, TRIAXIAL, PULSE, 48 OHMS, M17/19-RG25

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-17.

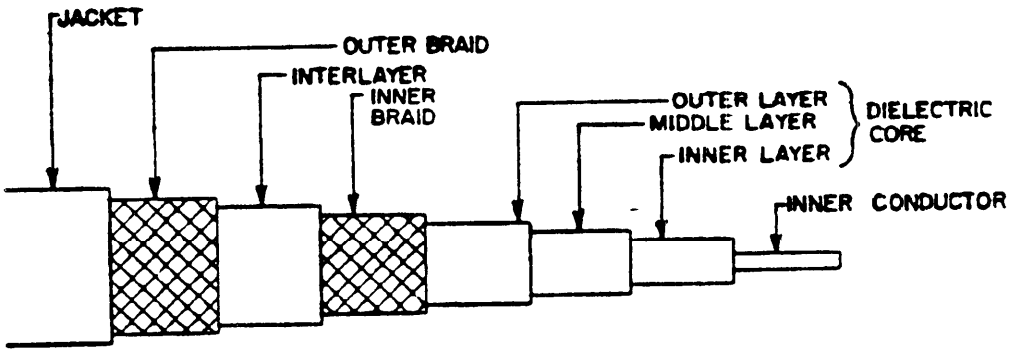


FIGURE 1. Configuration.

FSC 6145

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TABLE I. Description.

Components	Construction details
Inner conductor	Nineteen strands of tinned, copper wire, each strand .0117 inch diameter. Overall diameter: .0585 inch ±.0020.
Dielectric core Inner layer Middle layer Outer layer	Type E: Composite of three layers. Type H: Semiconducting synthetic rubber. Diameter: .090 inch ±.010. Type W: Insulating rubber. Diameter: .248 inch ±.010. Type Q: Insulating neoprene rubber. Colored, red. Diameter: .288 inch ±.010.
Inner braid	Single braid of AWG #34, tinned, copper wire. Diameter: .318 inch nominal. Coverage: 94.5%, nominal; Alternate: 94.3%, nominal Carriers: 16; Alternate: 24 Ends: 13; Alternate: 9 Picks/inch: 5.0 ±10%; Alternate: 6.0 ±10%
Interlayer	Type C-1: Braid of cotton tapes. Diameter: .353 inch nominal.
Outer braid	Single braid of AWG #34, tinned, copper wire Diameter: .398 inch maximum. Coverage: 95.3%, nominal; Alternate: 95.5%, nominal Carriers: 16; Alternate: 24 Ends: 15; Alternate: 11 Picks/inch: 5.0 ±10%; Alternate: 5.0 ±10%
Jacket	Type IV Diameter: .505 inch ±.015.

Engineering information:

Continuous working voltage: 7,500 Vrms, maximum.
Operating frequency: 1 MHz, maximum.
Velocity of propagation: 49.5 percent, nominal.
Operating temperature range: -55° to +90°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 0.482 ohms per 100 feet.
Elongation: 15% minimum.
Engineering notes: This cable useful in pulse low temperature applications (see connector per MIL-C-3607).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Adheson of conductors:
Inner conductor to core: 3 pounds minimum, 9 pounds maximum.
Aging stability: +98° ±2°C.
Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.
Cold bend: -40° ±2°C.

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Dimensional stability: +85 ±2°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Not applicable.
Weight: 22.5 pounds per 100 feet, maximum.

Electrical:
Spark test: 6,000 Vrms, +25%,-0%.
Voltage withstanding: 12,000 Vrms, minimum.
Insulation resistance: 500 megohms, minimum.
Corona extinction voltage: 10,000 Vrms, minimum.
Characteristic impedance: 48 ohms ±4.
Attenuation: 0.7 dB per 100 feet maximum at 1 MHz.
Structural return loss: Not applicable.
Capacitance: 50 pF per foot, maximum.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/19-RG25	RG25 per MIL-C-17/19

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-0716-2)

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